Large Language Models as Information Extractors in Healthcare

Leah Seseri
lseseri@ucsd.edu

Lewis Weng
lweng@ucsd.edu

Omid Alamdar
oalamdar@ucsd.edu

Aaron Boussina
aboussina@health.ucsd.edu

Abstract

Large Language Models (LLMs) show strong potential for clinical information extraction, however, electronic health records (EHR) often contain large amounts of redundant or irrelevant text that confuse the LLM making it struggle to reliably retrieve relevant information. Retrieval-augmented generation (RAG) can mitigate this challenge by reducing the amount of context provided to the LLM, although the most optimal approaches have not been explored. We introduce a benchmark for evaluating and comparing retrieval methods for EHR information extraction. Using the Severe Sepsis and Septic Shock Early Management Bundle (SEP-1) from the Centers for Medicare and Medicaid Services (CMS) specifications manual, we generated 100 synthetic needles representing positive qualifiers across five clinically meaningful elements. For each element, we randomly selected 10,000 patients from the MIMIC-III dataset (461,290 total clinical notes) and inserted one synthetic needle per patient to construct patient-level haystacks. We evaluated six retrieval strategies and prompted the LLM using top-10 retrieved passages. We assess the retrieval approaches against a baseline approach without RAG that uses all the information within the context limit of the model. LLM responses were generated using AWS Bedrock (DeepSeek-V3) and evaluated on classification and extraction tasks. SPLADE achieved the highest top-k recall, while BM25 achieved the lowest. Retrieval effectiveness varied by clinical element, with clinical trial queries achieving the highest recall and vasopressor queries the lowest. Baseline performance indicated that longer context and higher top-k retrieval reduced model accuracy, showcasing the need for retrieval. Overall, our results demonstrate a strong positive correlation between retrieval quality and LLM accuracy, emphasizing that developing and improving retrieval strategies can lead to more effective clinical information extraction.

Website: https://ucsd-dsc-capstone-b08-llm-team.github.io/llm-info-extractor-website/
Code: https://github.com/UCSD-DSC-Capstone-B08-LLM-Team/LLM_as_Info_Extractor

1 Introduction 3
2 Methods 4
3 Results 7
4 Discussion 10
5 Conclusion 11
References 12
Appendices A1

1 Introduction

Extracting information from medical charts is critical for patient care and healthcare operations. Medical documentation contains diagnoses, treatments, medications, laboratory results, and observations that directly influence medical decision making. Thus, it is significant that extracting this information from charts is not only done but achieved accurately. Moreover, not only does the information on medical charts matter, but there are numerous other tasks that depend on this information. Extracted chart data can be used for direct patient care and populating clinical registries. Such registries include providing information to advance medical research or reporting data to regulatory bodies so that guidelines and standards are enforced. One example of a regulatory body that data needs to be reported to is the Centers for Medicare and Medicaid Services (CMS). CMS oversees programs such as Medicare, Medicaid, and other reimbursement policies designed to improve access and equity of healthcare systems (Dasari et al. 2024). They run reporting programs to evaluate hospital performance which hospitals provide by reviewing medical charts and extracting clinical data. However, this process can be time consuming and often requires the effort of medical staff making quality reporting a costly activity (Boussina et al. 2024). Therefore, medical chart data is not merely stored for documentation. Extracting this information is essential for hospital and patient care while also being the backbone for ancillary activities.

Large language models have been used to extract unstructured information from electronic health records (EHR). However, it is well known that large language models struggle to extract long contexts effectively. An article from Chroma found that having larger having context windows with irrelevant context often ends up in worse performance for an LLM (Hong, Troynikov and Huber 2025). As additional context was added, particularly within the middle of an LLM's context, researchers often found the LLMs performance declined substantially.

Another limitation that comes with using LLM's to extract information is that oftentimes the LLM produces results that are inaccurate or unrelated to the task at hand. One study attempted to store numeric information from LLMs into structured records and used several heruistic-based approaches to reduce hallucinations (Adam et al. 2025). These approaches included using a regex-based cross-checking scheme, removing values that were used as examples in few-shot prompting, and removing values outside of a predefined range of plausible values. With these techniques, LLMs were as effective at parsing numeric values as carefully constructed, specific regex expressions, though taking substantially less time.

Recent work demonstrated that LLMs can successfully extract clinically important information from medical text. In a 2024 study, multiple large language models were evaluated on synthetic clinical notes with all models performing well, and with Claude achieving the best performance (Ntinopoulos et al. 2025). One study uses the annotation of manual reviewers as the ground truth, and found that Chat-GPT performed perfect in some categories, such as detecting smoking and not detecting cancer when not present, but performed worse (58%) in other categories, such as identifying Family History of Heart Disease (Bhagat et al. 2024). Other studies include humans working alongside LLM's to annotate health data to generate ground truth labels. They found that LLM's helped reviewers work on average 54% faster.

In addition, when the LLM was confident in its answer, it achieved a 96.3% accuracy, showing that the increase in speed did not come with at a notable loss (Goel et al. 2023). These findings indicate that LLMs have are capable of clinical information extraction, however, the effectiveness of these approaches depend on retrieving relevant context from large clinical records.

Retrieval-augmented generation (RAG) is a common approach used to improve LLM accuracy by retrieving relevant information before model inference. However, evaluation of retrieval strategies and the approaches that may be most optimal for EHR data extraction remain underexplored. To address this gap, this work introduces a benchmark for evaluating information retrieval approaches in medicine by covering a diverse range of methods, metrics, and clinical scenarios.

We use the MIMIC-III Clinical Database which contains de-identified EHR data for around 46,000 patients from the critical care units of Beth Israel Deaconess Medical Center in Boston between 2001 and 2012 (Johnson, Pollard and Mark 2016). It includes structured data such as demographics, vital signs, lab results, medications, and diagnostic codes, in which identifying information were de-identified, but we focus on the large collection of unstructured clinical notes present in the data. The NOTEEVENTS table contains a large amount of free text notes that vary in length, structure, and purpose, often written in domain specific language with variability in phrasing and abbreviations. We used the TEXT column from the NOTEEVENTS table, which contains a wide range of unstructured clinical documentation, including nursing notes, physician notes, ECG reports, radiology reports, and discharge summaries (Johnson, Pollard and Mark 2016).

This work provides (1) a benchmark for evaluating retrieval approaches on clinical notes, (2) a systematic comparison of retrieval strategies, and (3) insights into which approaches perform best for EHR information extraction.

2 Methods

2.1 Data Preparation

To evaluate LLM performance in a controlled needle-in-a-haystack setting, we utilized version 5.18a of the Centers for Medicare and Medicaid Services (CMS) Specifications Manual for National Hospital Inpatient Quality Measures. In particular, we used the SEP-1 1b-Alpha Data Dictionary (AlphaDD) abstraction specifications to define clinically meaningful queries that served as inputs to the retrieval step for identifying relevant passages (Centers for Medicare & Medicaid Services 2025).

We constructed queries representing positive qualifiers based on five elements described in the manual: Administrative Contraindication to Care–Severe Sepsis, Directive for Comfort Care or Palliative Care–Severe Sepsis, Clinical Trial–Severe Sepsis, Severe Sepsis Present, and Vasopressor Administration–Severe Sepsis. Synthetic ”needle” statements were generated from these queries using template clinical note patterns described in the CMS docu-

mentation. This approach ensured semantic diversity while maintaining clinical validity. Each generated needle satisfied the abstraction logic defined in the manual and provided a definitive response to the associated query, allowing the needles to serve as ground truth evidence for evaluation.

For each patient, all notes contained in the TEXT field were aggregated to create a patient level document representing the clinical "haystack". One synthetic needle corresponding to one of the five clinical elements was then randomly inserted into each patient's aggregated notes. We randomly selected 10,000 patients from the dataset and inserted a needle into each patient for all five elements. This procedure produced long context documents that simulate the review of a full patient chart. This experimental design consisting of a patient level haystack, a query derived from SEP-1 abstraction logic, and a known piece of supporting evidence which is the needle, enabled systematic evaluation of retrieval performance and downstream LLM tasks, including classification and information extraction.

2.2 Retrieval Methods

The retrieval approaches evaluated in this study included Best Matching 25 (BM25), Facebook AI Similarity Search (FAISS), FAISS using Maximal Marginal Relevance (MMR), semantic chunking, Sparse Lexical and Dense (SPLADE), and a hybrid retrieval approach.

BM25 is a widely used lexical retrieval algorithm that ranks documents based on term frequency and inverse document frequency (IDF) (Robertson and Zaragoza 2009). While BM25 is effective for exact keyword matching, it often struggles to capture semantic relationships when queries and documents use different terminology.

We also evaluated dense retrieval using FAISS, a vector search library that enables similarity search over high dimensional embeddings (Douze et al. 2025). Text passages were converted into vector representations using sentence-transformer embeddings, and similarity between query and passage vectors was computed using both cosine similarity and Euclidean distance. To further improve retrieval diversity, we implemented MMR with FAISS (Carbonell and Goldstein 1998). MMR selects passages that maximize relevance to the query while minimizing redundancy among retrieved passages, thereby increasing coverage of distinct clinical evidence.

In semantic chunking, the haystack was segmented into contextually coherent units. This method detects semantic shifts within the text and groups related sentences into meaningful segments before computing embeddings. By preserving semantic coherence within each chunk, this approach can improve retrieval quality by ensuring that related clinical evidence remains grouped together.

SPLADE is a sparse retrieval model that expands queries and documents into weighted lexical representations using a transformer-based encoder(Formal, Piwowarski and Clinchant 2021). Unlike traditional lexical methods, SPLADE assigns weights to terms based on contextual importance, enabling semantic matching while retaining the efficiency of inverted index search structures. This approach improves lexical recall by allowing retrieval of semantically related terms and synonyms.

Finally, we implemented a hybrid retrieval approach that combines BM25 and FAISS retrieval scores (Hakdağlı 2024). Passage scores from each method were combined using equal weighting, allowing the hybrid approach to leverage the complementary strengths of lexical and semantic retrieval.

To evaluate the necessity of retrieval prior to LLM prompting, we also implemented a baseline approach. The full-context baseline is that the entire patient haystack was provided directly to the LLM without retrieval. Thus, instead of entering the retrieved passages, the entire haystack acts as the passage. Due to model token limitations, this context was truncated to the max token input allowed. The same context-length constraint was applied to all retrieval based prompts to ensure fair comparison across methods.

2.3 LLM Prompting and Answer Generation

All LLM inference was conducted using AWS Bedrock, which provides a secure environment compliant with HIPAA and institutional data use agreements (DUA).

We experimented with several models available through Bedrock, including Claude 3 and DeepSeek-V3, and ultimately selected DeepSeek-V3 for the primary experiments due to its favorable balance between cost and performance (DeepSeek-AI et al. 2025). In the MedHELM benchmark conducted on multiple LLM's, including versions of Claude, GPT, Gemini, and Llama, Deep Seek achieved the highest win-rate of 0.66, which represents the proportion of pairwise comparisons where each model achieved superior performance across all 35 (Bedi et al. 2025).

For each retrieval strategy, the top-k passages returned by the retriever were used to construct the prompt context. The prompt structure followed a consistent format in which the instruction and the retrieved passages were provided first, followed by the query.

The LLM was tasked with performing two types of operations: classification and extraction. Classification required the model to determine whether the provided passages contained sufficient information to answer the query. Information extraction required the model to identify and extract the specific clinical evidence corresponding to the query. In the classification task, a positive ("Yes") response indicates that the model identified evidence relevant to the query. In the extraction task, baseline and retrieval method outputs were compared against the ground-truth synthetic needles to determine whether the correct evidence had been retrieved and extracted.

2.4 Evaluation

Model outputs were evaluated by comparing responses to the known synthetic needles embedded within each patient document. Retrieval effectiveness was assessed by measuring whether the inserted needle appeared within the retrieved passages. LLM performance was evaluated using classification and extraction accuracy across retrieval strategies and clinical elements.

3 Results

Across all retrieval approaches, SPLADE consistently achieved the highest top-k recall whereas BM25 consistently exhibited the lowest recall across the five clinical elements. FAISS and semantic chunking demonstrated intermediate performance, outperforming BM25, but not achieving the highest recall. Hybrid retrieval provided more consistent performance, surpassing FAISS and BM25 alone when both individual methods performed well. However, in elements where BM25 underperformed relative to FAISS, hybrid retrieval demonstrated limited improvement. These results suggest that combining lexical and semantic approaches can enhance retrieval effectiveness in clinical contexts, particularly when both methods individually perform adequately. Furthermore, FAISS augmented with MMR outperformed FAISS alone, as MMR improved recall by reducing redundancy among retrieved passages while maintaining relevance. Figure 1 presents the top-k recall for each retrieval method across the five clinical elements, while Table 1 summarizes the numeric results.

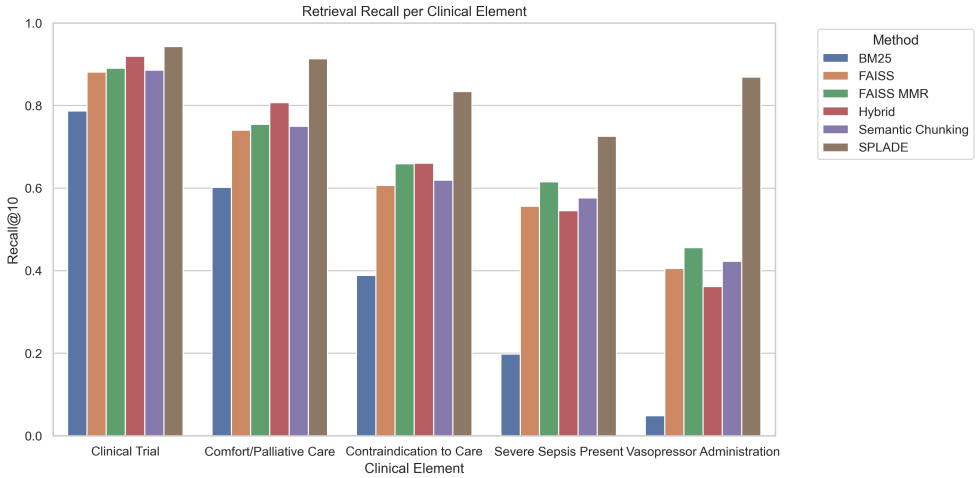

Figure 1: Top-k recall for each retrieval method across the five clinical elements. SPLADE consistently achieves the highest recall across elements, while BM25 shows the lowest recall overall. Retrieval performance varies by clinical element, with clinical trial achieving the highest recall and vasopressor the lowest.

Table 1: Top-k recall for each retrieval method across the five clinical elements.

Clinical Element	BM25	FAISS	FAISS+MMR	Hybrid	SPLADE	Semantic Chunking
Clinical Trial	0.7865	0.8812	0.8902	0.9192	0.9432	0.8855
Comfort Care	0.6015	0.7405	0.7544	0.8071	0.9133	0.7497
Contraindication Care	0.3883	0.6063	0.6589	0.6600	0.8337	0.6190
Severe Sepsis	0.1982	0.5563	0.6152	0.5451	0.7253	0.5762
Vasopressor	0.0488	0.4050	0.4557	0.3612	0.8692	0.4230

Analyzing performance across the five clinical elements revealed that retrieval effectiveness varied by element. Recall at k was highest for the clinical trial related haystack and needles, with the directive for comfort care also demonstrating high recall. In contrast, the vasopressor element exhibited the lowest recall across all retrieval methods. Recall for severe sepsis

and administrative contraindication to care was intermediate, outperforming vasopressor but underperforming relative to clinical trial and comfort care. The variance in our results indicates that retrieval is dependent on factors other than the data surrounding the needle.

For downstream tasks involving the LLM, we focused on classification and extraction. For extraction tasks, we evaluated performance relative to a baseline that always included the complete haystack, achieving 100% needle extraction, as expected. Retrieval based strategies exhibited variable success, with SPLADE consistently outperforming other methods. Across all elements, the clinical trial element achieved the highest retrieval success.

For classification tasks, the LLM generally performed better when provided with retrieved passages than when provided with the baseline haystack. Figure 2 shows the improvement in LLM accuracy for each retrieval method compared to the baseline approach. Among re-

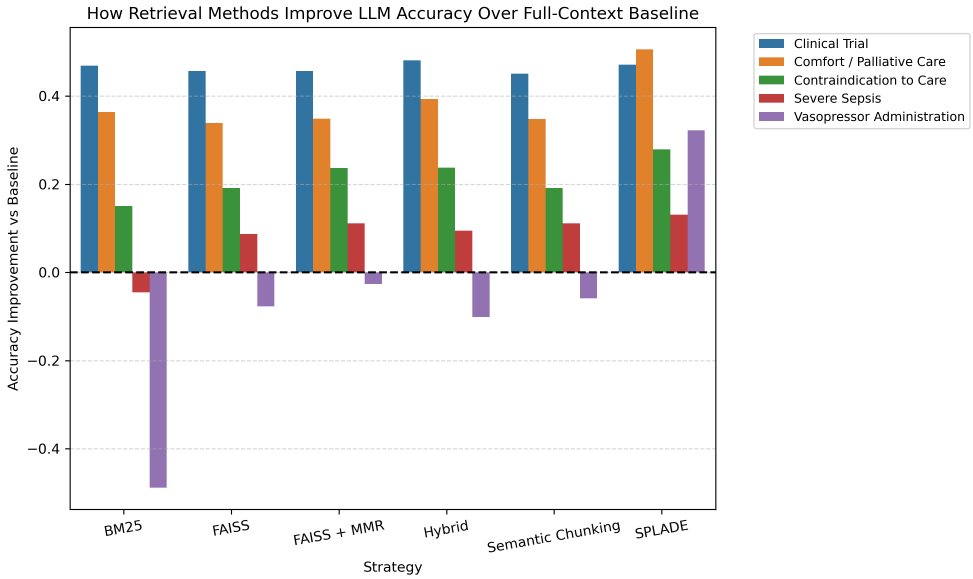

Figure 2: Improvement in LLM accuracy when using retrieval based methods compared to the baseline across clinical elements. Most retrieval strategies improve LLM performance; however, BM25 retrieval for the vasopressor element shows little or no improvement.

trieval strategies, SPLADE led to the highest classification accuracy, followed by FAISS with MMR. These findings indicate that LLMs perform more accurately when supplies with relevant passages rather than with large unfiltered documents. Performance was highest for the clinical trial and comfort care elements, whereas classification accuracy was lower for contraindications to care and severe sepsis. Interestingly, vasopressor emerges as an outlier: the LLM performed well across all methods, including the baseline. Within this element, SPLADE derived passages yielded the highest LLM accuracy, whereas BM25 derived passages yielded the lowest across all elements and retrieval approaches. Figure 3 illustrates the LLM classification and extraction accuracy for each retrieval method across the clinical elements, and provides a detailed numeric summary including baseline performance.

We also investigated the effect of context length and top-k retrieval. Increasing context decreased LLM accuracy, likely due to the inclusion of extraneous information that distracted

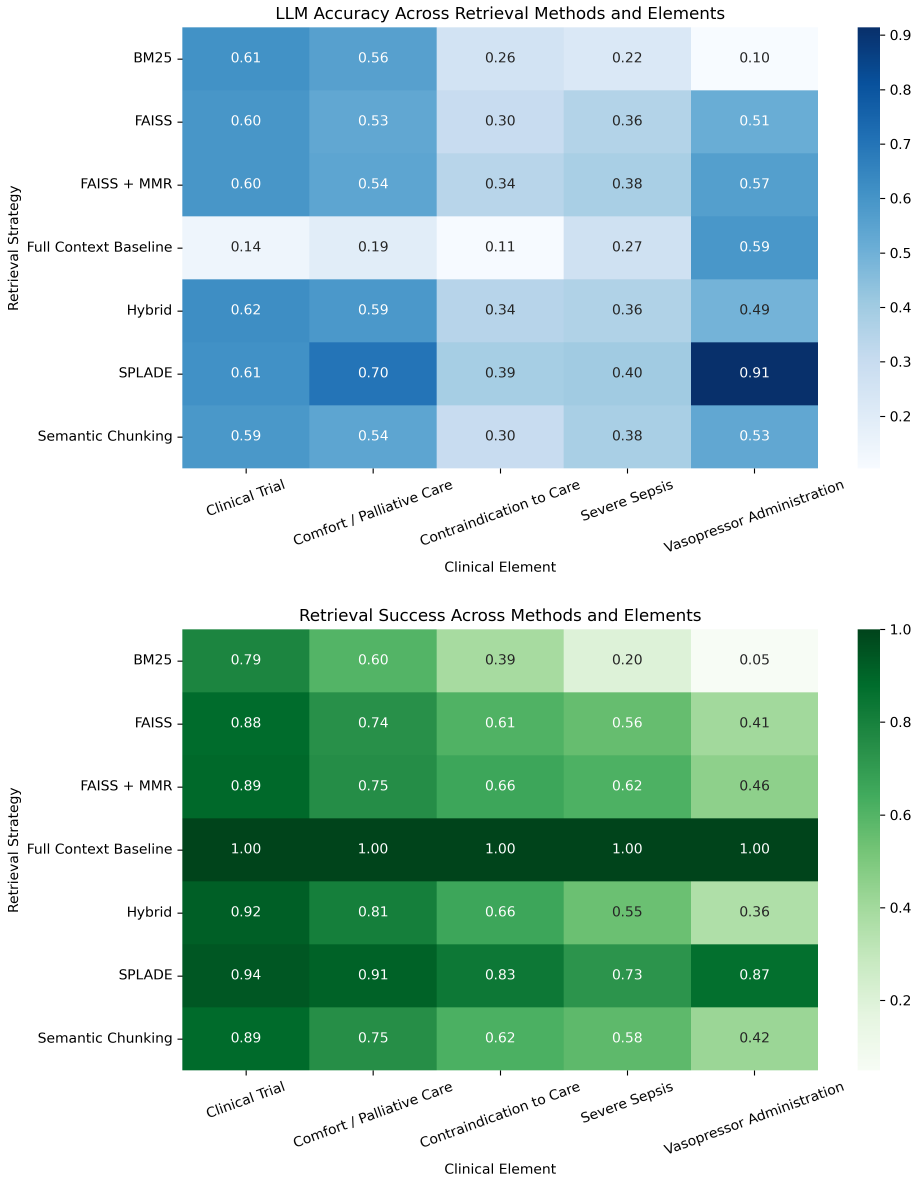

Figure 3: LLM performance across retrieval methods and clinical elements. The top panel shows classification accuracy and the bottom panel shows extraction accuracy for each retrieval strategy.

the model. Similarly, retrieving more passages increased the likelihood of including the needle but introduced additional irrelevant information, slightly reducing LLM classification accuracy. Based on these findings, we selected the top-10 most relevant passages for each retrieval methods, balancing retrieval success with accurate LLM performance.

Overall, there is a strong positive correlation between retrieval success and LLM accuracy, as illustrated in Figure 4. High recall retrieval consistently led to more accurate LLM responses, demonstrating that LLMs can serve as effective information extractors when provided with relevant, high quality passages.

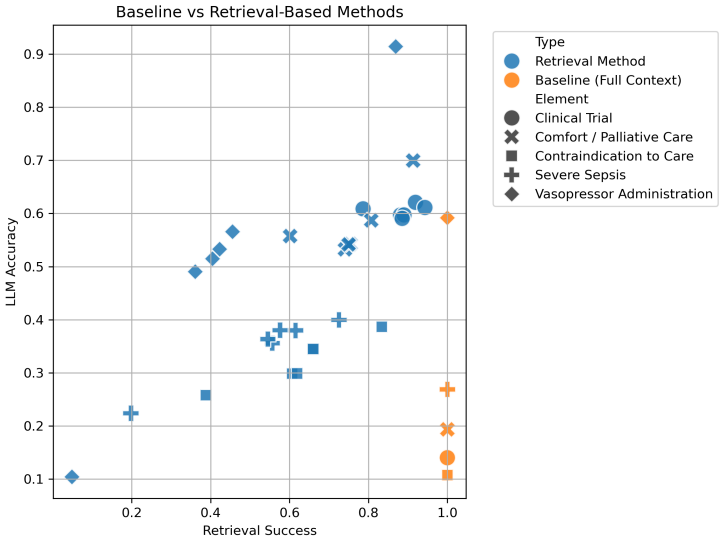

Figure 4: Relationship between retrieval recall and LLM classification accuracy across retrieval methods. Higher retrieval recall generally corresponds to higher LLM accuracy. Although the baseline includes the entire document and therefore achieves 100% retrieval recall, the LLM performs worse due to the large amount of irrelevant context.

4 Discussion

Our results highlight that retrieval quality is a critical factor in maintaining overall pipeline accuracy. Effective retrieval substantially mitigates the challenges that are associated with extracting information from long clinical documents. When relevant passages are successfully retrieved, LLMs are better able to identify the required information. Conversely, when retrieval returns suboptimal passages, even highly capable LLMs struggle to produce accurate responses. These findings reinforce the importance of retrieval and support prior work demonstrating that retrieval quality often serves as the primary bottleneck in RAG systems.

Several challenges inherent to long clinical documentation contribute to these difficulties. As context length increases, LLM performance can degrade due to phenomena such as the lost-in-the-middle effect and the presence of distractors. Although retrieval reduces the overall context size presented to the model, it does not completely eliminate the confusion

introduced by semantically similar but irrelevant passages. This challenge is particularly evident in datasets such as MIMIC-III, where many clinical notes share similar structures, terminology, and patient conditions. As a result, distinguishing clinically relevant information from surrounding text remains nontrivial.

Our approach of generating synthetic needles enabled us to evaluate retrieval and LLM performance in a controlled setting while constructing a realistic evaluation framework that simultaneously avoids the privacy risks associated with handling sensitive patient information. We evaluated five distinct clinical elements to capture a broader range of information types and reflect the diversity of real world medical documentation. Lexical retrieval methods performed well when query phrasing closely matched the text, semantic approaches were more effective at identifying underlying meaning, and hybrid methods showed promise by combining the complementary strengths of both. Prioritizing the most relevant passages within the retrieved context helped mitigate the lost-in-the-middle effect. Together, these experiments establish a unique benchmark for information retrieval for EHR data.

Despite these advantages, our approach has several limitations. Synthetic text cannot fully capture the complexity and variability present in real clinical documentation, particularly in cases involving adverse events or ambiguous clinical narratives. Furthermore, random needle insertion may produce patterns that are more structured than those observed in real world workflows.

Future work should therefore incorporate real clinical scenarios or structured labels to approximate real clinical medical workflows more closely. Our benchmark demonstrates that existing retrieval strategies do not yet achieve the performance required for deployment in the medical setting, revealing areas where futher optimization is necessary. By providing a standardized and reusable framework, this benchmark enables researchers to systematically evaluate and develop retrieval strategies under realistic conditions while maintaining patient privacy. Since retrieval improvements do not require larger models or access to sensitive training data, these strategies offer a scalable and practical pathway for deploying LLMs as information extractors in healthcare. Overall, the benchmark itself provides a valuable resource for the development of more reliable and clinically deployable information extraction systems.

5 Conclusion

This study highlights a central insight: the effectiveness of LLM based clinical information extraction depends strongly on the quality of the retrieved context. Even highly capable language models struggle when relevant passages are not retrieved, whereas strong retrieval substantially improves downstream task performance. Using needle-in-a-haystack evaluations across thousands of clinical notes, we systematically examined how different retrieval strategies influence LLM accuracy in clinical information extraction tasks. In certain contexts, LLM's show promising results and have the potential to become employed in industry after further testing, speeding up the medical process of reading medical charts.

Future work will focus on extending this evaluation through additional quantitative analyses of retrieval and LLM performance, exploring more advanced retrieval pipelines, and evaluating these approaches on longer and more complex clinical documents. We also plan to examine additional datasets to determine whether the observed trends generalize beyond the MIMIC-III corpus.

Overall, this work demonstrates both the promise and the challenges of applying LLMs to clinical text extraction. When combined with high quality retrieval strategies, LLMs have the potential to serve as powerful tools for identifying clinically relevant information within large volumes of medical documentation. Continued research in retrieval optimization and retrieval aware prompting may further improve the reliability and effectiveness of these systems, ultimately supporting clinicians by reducing documentation burden and improving access to critical patient information.

References

Adam, Hammaad, Junjing Lin, Jianchang Lin, Hillary Keenan, Ashia Wilson, and Marzyeh Ghassemi. 2025. "Clinical Information Extraction with Large Language Models: A Case Study on Organ Procurement." In *AMIA Annual Symposium Proceedings*.

Bedi, Suhana, Hejie Cui, Miguel Fuentes, Alyssa Unell, Michael Wornow, Juan M. Banda, Nikesh Kotecha, Timothy Keyes, Yifan Mai, Mert Oez, Hao Qiu, Shrey Jain, Leonardo Schettini, Mehr Kashyap, Jason Alan Fries, Akshay Swaminathan, Philip Chung, Fateme Nateghi, Asad Aali, Ashwin Nayak, Shivam Vedak, Sneha S. Jain, Birju Patel, Oluseyi Fayanju, Shreya Shah, Ethan Goh, Dong han Yao, Brian Soetikno, Eduardo Reis, Sergios Gatidis, Vasu Divi, Robson Capasso, Rachna Saralkar, Chia-Chun Chiang, Jenelle Jindal, Tho Pham, Faraz Ghoddusi, Steven Lin, Albert S. Chiou, Christy Hong, Mohana Roy, Michael F. Gensheimer, Hinesh Patel, Kevin Schulman, Dev Dash, Danton Char, Lance Downing, Francois Grolleau, Kameron Black, Bethel Mieso, Aydin Zahedivash, Wen wai Yim, Harshita Sharma, Tony Lee, Hannah Kirsch, Jennifer Lee, Nerissa Ambers, Carlene Lugtu, Aditya Sharma, Bilal Mawji, Alex Alekseyev, Vicky Zhou, Vikas Kakkar, Jarrod Helzer, Anurang Revri, Yair Bannett, Roxana Daneshjou, Jonathan Chen, Emily Alsentzer, Keith Morse, Nirmal Ravi, Nima Aghaeepour, Vanessa Kennedy, Akshay Chaudhari, Thomas Wang, Sanmi Koyejo, Matthew P. Lungren, Eric Horvitz, Percy Liang, Mike Pfeffer, and Nigam H. Shah. 2025. "MedHELM: Holistic Evaluation of Large Language Models for Medical Tasks." [Link]

Bhagat, Navya et al. 2024. "Large Language Models for Efficient Medical Information Extraction." *AMIA Joint Summits on Translational Science proceedings. AMIA Joint Summits on Translational Science* 2024: 509–514

Boussina, Aaron, Rishivardhan Krishnamoorthy, Kimberly Quintero, Shreyansh Joshi, Gabriel Wardi, Hayden Pour, Nicholas Hilbert, Atul Malhotra, Michael Hogarth, Amy M Sitapati et al. 2024. "Large language models for more efficient reporting of hospital quality measures." *Nejm ai* 1 (11), p. AIcs2400420

Carbonell, Jaime, and Jade Goldstein. 1998. "The use of MMR, diversity-based reranking for reordering documents and producing summaries." In *Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*. New York, NY, USA Association for Computing Machinery. [Link]

Centers for Medicare & Medicaid Services. 2025. *SEP-1 Early Management Bundle for Severe Sepsis/Septic Shock: 1b Alpha Data Dictionary*. Centers for Medicare & Medicaid Services, Baltimore, MD. [Link]

Dasari, Shobha, Raihana Mehreen, Kristin Baker Spohn, and Andrey Ostrovsky. 2024. "Opportunities for CMS to improve healthcare access and equity through advancing technology-enabled startups and digital health innovations." *NPJ digital medicine* 7 (1), p. 23

DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda

Lu, Chenggang Zhao, Chengqi Deng, Chenyu Zhang, Chong Ruan, Damai Dai, Daya Guo, Dejian Yang, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Han Bao, Hanwei Xu, Haocheng Wang, Haowei Zhang, Honghui Ding, Huajian Xin, Huazuo Gao, Hui Li, Hui Qu, J. L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jiawei Wang, Jin Chen, Jingchang Chen, Jingyang Yuan, Junjie Qiu, Junlong Li, Junxiao Song, Kai Dong, Kai Hu, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Litong Wang, Liyue Zhang, Meng Li, Miaojun Wang, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang, Peng Zhang, Qiancheng Wang, Qihao Zhu, Qinyu Chen, Qiushi Du, R. J. Chen, R. L. Jin, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, Runxin Xu, Ruoyu Zhang, Ruyi Chen, S. S. Li, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shaoqing Wu, Shengfeng Ye, Shengfeng Ye, Shirong Ma, Shiyu Wang, Shuang Zhou, Shuiping Yu, Shunfeng Zhou, Shuting Pan, T. Wang, Tao Yun, Tian Pei, Tianyu Sun, W. L. Xiao, Wangding Zeng, Wanjia Zhao, Wei An, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, X. Q. Li, Xiangyue Jin, Xianzu Wang, Xiao Bi, Xiaodong Liu, Xiaohan Wang, Xiaojin Shen, Xiaokang Chen, Xiaokang Zhang, Xiaosha Chen, Xiaotao Nie, Xiaowen Sun, Xiaoxiang Wang, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xingkai Yu, Xinnan Song, Xinxia Shan, Xinyi Zhou, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, Y. K. Li, Y. Q. Wang, Y. X. Wei, Y. X. Zhu, Yang Zhang, Yanhong Xu, Yanhong Xu, Yanping Huang, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Li, Yaohui Wang, Yi Yu, Yi Zheng, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Ying Tang, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yu Wu, Yuan Ou, Yuchen Zhu, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yukun Zha, Yunfan Xiong, Yunxian Ma, Yuting Yan, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Z. F. Wu, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhen Huang, Zhen Zhang, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhibin Gou, Zhicheng Ma, Zhigang Yan, Zhihong Shao, Zhipeng Xu, Zhiyu Wu, Zhongyu Zhang, Zhuoshu Li, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Ziyi Gao, and Zizheng Pan. 2025. "DeepSeek-V3 Technical Report." [Link]

Douze, Matthijs, Alexandr Guzhva, Chengqi Deng, Jeff Johnson, Gergely Szilvasy, Pierre-Emmanuel Mazaré, Maria Lomeli, Lucas Hosseini, and Hervé Jégou. 2025. "The Faiss library." [Link]

Formal, Thibault, Benjamin Piwowarski, and Stéphane Clinchant. 2021. "SPLADE: Sparse Lexical and Expansion Model for First Stage Ranking." In *Proceedings of the 44th International ACM SIGIR Conference on Research and Development in Information Retrieval*. New York, NY, USA Association for Computing Machinery. [Link]

Goel, Akshay, Almog Gueta, Omry Gilon, Chang Liu, Sofia Erell, Lan Huong Nguyen, Xiaohong Hao, Bolous Jaber, Shashir Reddy, Rupesh Kartha, Jean Steiner, Itay Laish, and Amir Feder. 2023. "LLMs Accelerate Annotation for Medical Information Extraction."

In *AMIA Annual Symposium Proceedings*.

Hakdağlı, Özlem. 2024. "Hybrid Question-Answering System: A FAISS and BM25 Approach for Extracting Information from Technical Document." *Orclever Proceedings of Research and Development* 5 (1), p. 226–237. [Link]

Hong, Kelly, Anton Troynikov, and Jeff Huber. 2025. "Context rot: How increasing input tokens impacts LLM performance." Technical Report, Chroma. [Link]

Johnson, Alistair, Tom Pollard, and Roger Mark. 2016. "MIMIC-III Clinical Database." *PhysioNet*. [Link]

Ntinopoulos, Vasileios, Hector Rodriguez Cetina Biefer, Igor Tudorache, Nestoras Papadopoulos, Dragan Odavic, Petar Risteski, Achim Haeussler, and Omer Dzemali. 2025. "Large language models for data extraction from unstructured and semi-structured electronic health records: a multiple model performance evaluation." *BMJ health & care informatics* 32 (1), p. e101139

Robertson, Stephen, and Hugo Zaragoza. 2009. "The Probabilistic Relevance Framework: BM25 and Beyond." *Found. Trends Inf. Retr.* 3 (4), p. 333–389. [Link]

Appendices

A.1 Related Works A1

A.1 Related Works

This appendix provides additional context and implementation details supporting the experiments described in this report. During Quarter 2, the project expanded significantly beyond the initial prototype developed in Quarter 1. The updated system introduced a more comprehensive retrieval and evaluation pipeline designed to operate on patient-level clinical note collections derived from the MIMIC-III NOTEEVENTS dataset. Multiple clinical notes for each patient were aggregated to form long-context documents that simulate realistic electronic health record (EHR) charts. Synthetic evidence statements, referred to as “needles”, were inserted into these documents to create controlled needle-in-a-haystack scenarios for evaluating retrieval and language model performance. Retrieval strategies implemented in the expanded pipeline included lexical search using BM25, dense vector retrieval using FAISS with cosine similarity and Euclidean distance, FAISS with maximal marginal relevance (MMR) to improve diversity among retrieved passages, semantic chunking for context-aware document segmentation, SPLADE sparse retrieval for transformer-based lexical expansion, and hybrid retrieval approaches combining lexical and semantic similarity signals. These retrieval methods were evaluated based on their ability to surface the relevant evidence within the top-k retrieved passages. The retrieved passages were then used to construct prompts for large language models deployed through AWS Bedrock, primarily using the DeepSeek-V3 model, to perform classification and extraction tasks related to clinically meaningful SEP-1 abstraction queries. The experimental pipeline processed prompts stored in CSV files and executed Bedrock inference requests in parallel using a multithreaded evaluation script, enabling efficient experimentation across thousands of patient records. Model outputs were stored alongside the original prompts to allow downstream evaluation, visualization, and comparison across retrieval strategies. These additional implementation details, retrieval configurations, and evaluation artifacts provide reproducibility for the experimental setup presented in this report.

This an example of a prompt provided to the LLM: ”In this note, is there an explicit positive qualifier for an infection? Do not infer or take a likely guess. Answer Y only if the infectious condition is clearly stated in the note. Do not use documentation of viral, fungal, or parasitic infections. The following is a list of conditions that are considered infections (this is not an all-inclusive list.): EXAMPLES: Abscess, Acute abdomen, Acute abdominal infection, Blood stream, catheter infection, Bone/joint infection, C. difficile (C-diff), Chronic obstructive pulmonary disease (COPD) acute exacerbation, Endo-

carditis, Gangrene, Implantable device infection, Infection, Infectious, Meningitis, Necrosis, Necrotic/ischemic/infarcted bowel, Pelvic Inflammatory Disease, Perforated bowel, Pneumonia, Empyema, Purulence/pus, Sepsis, Septic, Skin/soft tissue infection, Suspect infection source unknown, Urosepsis, Urinary tract infection, Wound infection. EXAMPLES THAT AREN'T INFECTIONS: Fever, Diarrhea. POSITIVE QUALIFIERS: Possible, Rule out (r/o), Suspected, Likely, Probable, Differential diagnosis, Suspicious for, Concern for, Suggestive of, Presumed. NEGATIVE QUALIFIERS: Impending, Unlikely, Doubt, Ruled out, Less likely, Questionable."